

Match definition and word:

Ion	Alkali Metals	Atomic radius	Taxonomy	Octet rule
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- 1) Atoms want to have either 8 or 0 electrons in their outermost shell.
 - 2) The first group in the periodic table
 - 3) Trend on the periodic table in which Fr is the high end and He is the low end.
 - 4) Study of relatedness and trends in biology
 - 5) Atoms with a differing number of electrons
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- 6) **Water** is a covalent bond, every water we see on the earth is made up of tiny little atoms.
 - a. Draw the **lewis structure** of water.

- b. Discuss whether water is polar or not, the electronegativity of H is 2.20 and that of O is 3.55.

- 7) **Life needs to be organized**, taxonomy is how we do that!
 - a. **Fill in** the missing parts of the taxonomic tree
Kingdom - P_____ - Class - O_____ - F_____ - Genus - S_____
 - b. **Describe** in which part of the taxonomy would animals be more related, in a **genus** or a **kingdom**.

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- c. What would be the scientific name of the species in the following family, genus and species: **wasin, dreepi, plopr**

8) Magnesium and oxygen form a particular bond.

- a. **Explain** which bond this is, and how you know.

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- b. **Draw** the dot and cross diagram of magnesium and oxygen, and draw arrows indicating the transfer of electrons.

- c. now **draw** the shell diagrams of each ion after this electron transferal.

9) Naming of elements can be tough sometimes. Luckily for you we can practice

- a. Name the following **ionic** compounds: **NaCl, Fe(II)O, Mg₃N₂**

b. **Name** the following covalent compounds: CO_4 , N_3F_2 , CH_4

c. **Name** the two forms of metallic bonds.

10) Sometimes (jk a lot of times) atoms meet and react to form compounds in a chemical equation. One of those is the formation of Chromium (III) Sulfide.

a. What type of compound is this?

b. **State** the charge of the chromium.

c. **Balance** the reaction for the formation of Chromium (III) Sulfide:
 $\text{Cr} + \text{S}_8 \rightarrow \text{Cr}_2\text{O}_3$

11) Certain bonding types have different properties. Below are some questions in which you elaborate on these properties.

a. **Explain** why ionic compounds cannot conduct electricity when solid, but are able to do so when dissolved or molten.

- b. Covalent bonds often have lower melting points than ionic compounds. **Explain** why this is.

- c. Metallic bonds are shiny and can conduct electricity. **Explain** how these two properties arise in metallic bonds.

- 12) Draw the **lewis structure** of the following compounds:

a. **NH₃**

b. **CO₂**

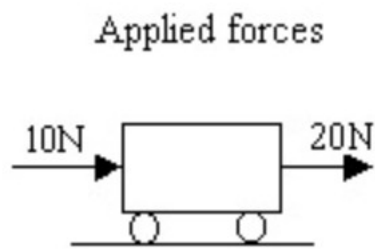
c. **C₃H₁₀**

- 13) **Forces** are AMAZING and can describe two things of the movement of an object:

- a. **Name these two** things: (Aka **state** what two things a force depicts)

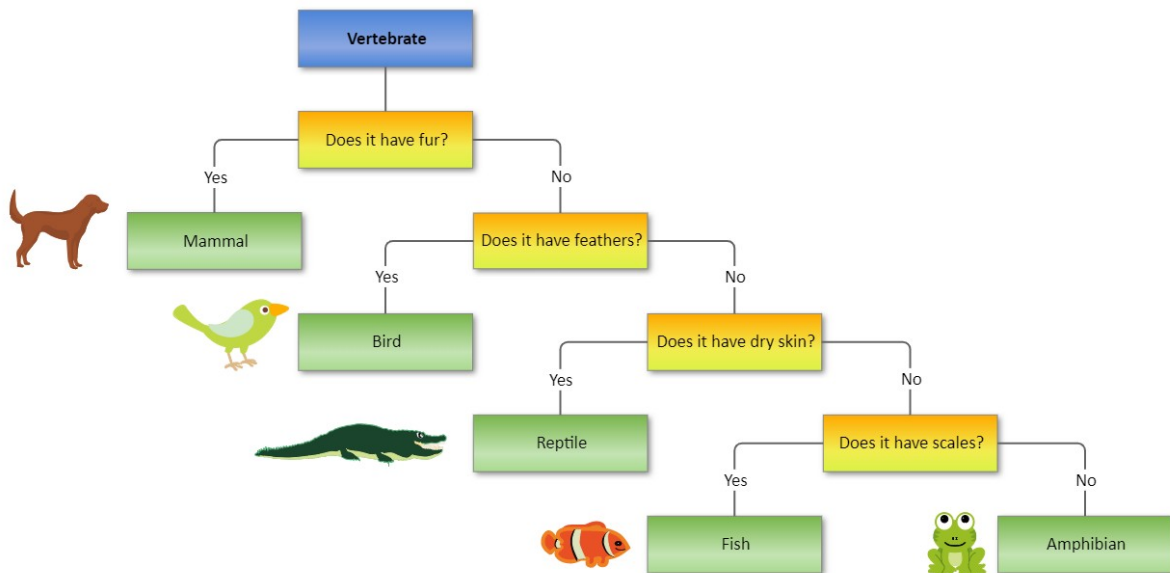
- b. **Draw** the forces acting on the box in the story below:
On earth, there is a box on wheels accelerating towards the right on a straight road.

- c. **Calculate** the net force in figure below.



14) Dichotomous keys

Dichotomous Key for Animals



a. **Explain** as what the animal to the right could be classified as

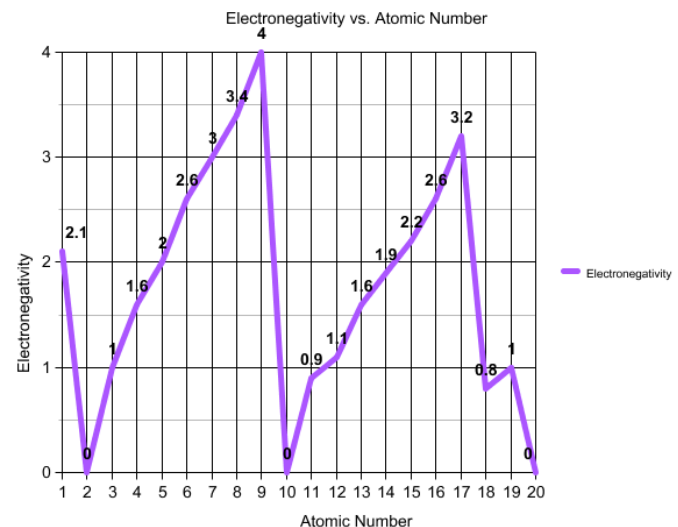


b. **Explain** how useful the dichotomous key above is in the following examples is:

- i. **Jimbo** lost his pet puffer fish in a heavy rainfall when he was taking it out on a walk. There are a lot of different pufferfish in this city, and Alex hopes that the dichotomous keys above can help the people in the city find **his pufferfish**.

15) **Graph exercise** the graph to the right shows the trend of electronegativity and the different atomic numbers.

- a. **Describe** the trend of electronegativity going down a period.



Explain the dips and peaks in the graph.
